

SEQUENCE ASSEMBLY PLAN

Research Methods in Biomedical Informatics

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OVERVIEW

Problem

“Create a program that takes as input the set of all next-generation sequencing reads identified in a sample and an initial query sequence and returns the largest sequence contig that can be constructed from the reads that contains the initial query sequence.”

Query sequence: **ZZZZZZZZ**

Sequencer reads:

ZZZWZYZY	YYYYYVYYY	XXXXZZZZ
ZZZZZZZZ	XXXXXXXXXX	ZZZZYYYYY
BBBBZBBYB	BBBBZBBYB	DDDDDDDDDD

Output contig: **XXXXXXXXXXZZZZZZZZZZ**YYYYYVYYY



“In a hole in the ground there lived a hobbit. Not a nasty, dirty, wet hole, filled with the ends of worms and an oozy smell, nor yet a dry, bare, sandy hole with nothing in it to sit down on or to eat: it was a hobbit-hole, and that means comfort.”

–*The Hobbit* [1]

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Chop into k length 'mers using a sliding window

$k = 5$

inaho, nahol, nahole, aholei, holein, ... itnot, tnota, notan, otana ...

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How do we work backwards?

Chop into k length 'mers using a sliding window

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A graph is a structure used to encode connections between things.

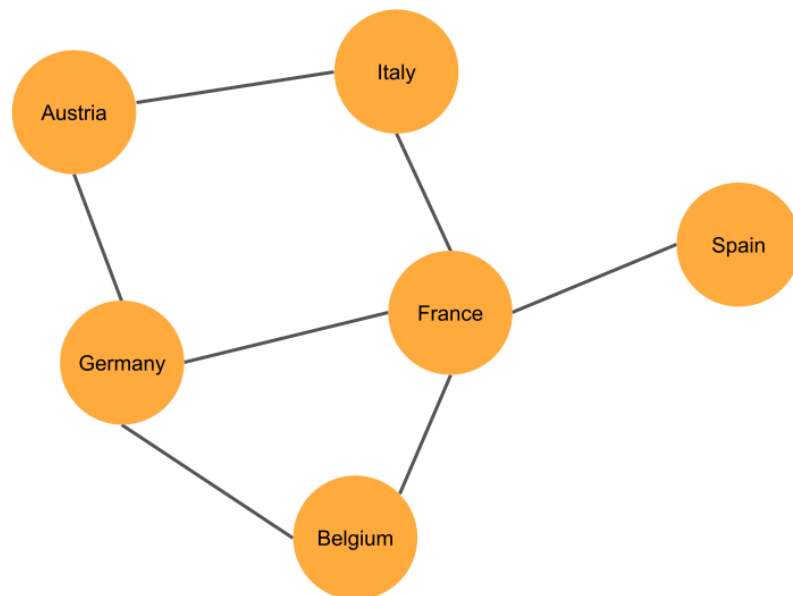


Figure 1: A simple graph used to represent how countries share borders.

I can use a graph to connect kmers that are similar to each other and reconstruct the original string by following each connection.

Using the previous example, I can plot the graph connecting each kmer to every other kmer that has an overlap.

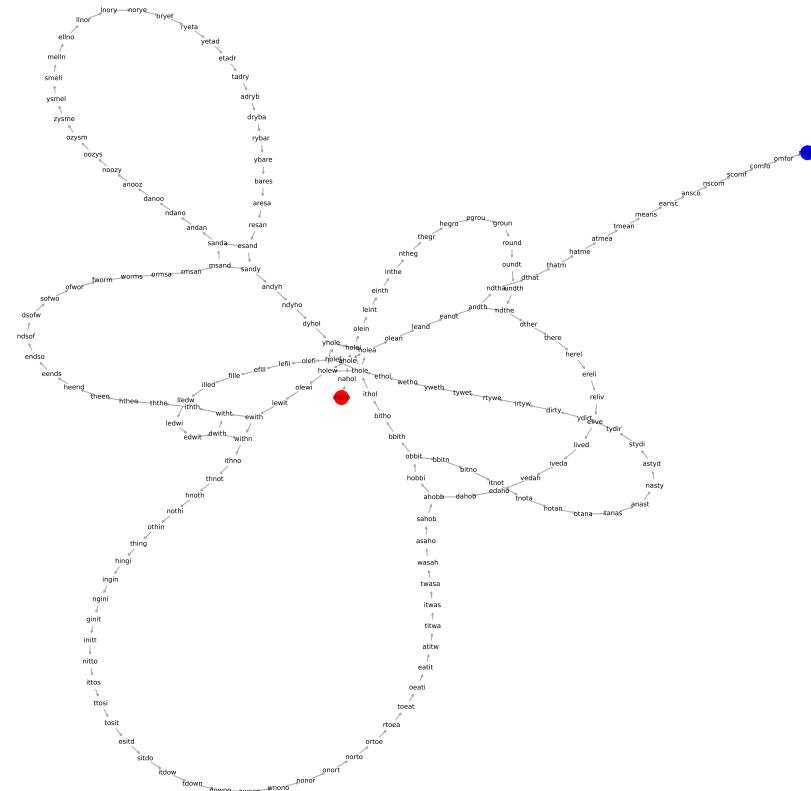
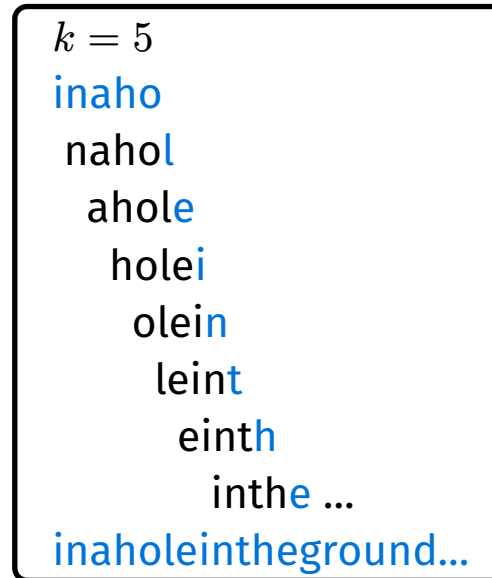


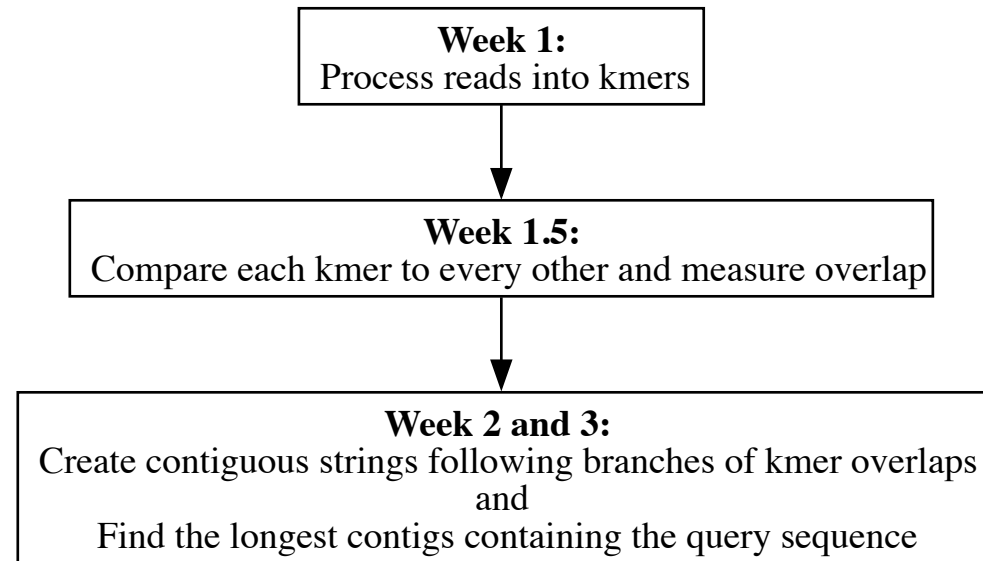
Figure 2: A graph of the 5-length kmers from the first sentence of *The Hobbit*.

This kind of graph is composed of kmers with $k-1$ overlapping characters.
AKA a De Bruijn graph:



Following each edge along the graph, the original sentence can be constructed using the last letter of each word.

DEVELOPMENT PLAN



- Make k-length strings (kmers) from reads.
- Record kmer frequencies.
- Filter out infrequent kmers below some threshold τ .
- Filter for unique kmers.
- Return list of unique kmers S .

```
let R be a list of strings composed of A, T, C, or G
let k be an integer
let tau be an integer
let results be an empty list
let counter be a dictionary

# trim each string to length k; keep track of string frequencies
for r in R:
    l = len(r)
    for i in range(0, l-k+1):
        r_sub = r[i:i+k]
        results += r_sub
        counter[r_sub] += 1

# save only unique and frequent strings
filtered = counter.keys()[counter.values() > tau]

return filtered
```

- Test how each kmer aligns with every other.
- Return an adjacency matrix where a value of 1 means the kmer at the row index matches the string of the column index.

```
let S be a list of k-length strings
let m be a square matrix of zeros with dimensions len(S)

for i in range(len(S)):
    for j in range(len(S)):
        if i does not equal j:
            si = S[i]
            sj = S[j]
            let si_sub be the substring of si from the second character to the last (2 to k)
            let sj_sub be the substring of sj from the first character to the second last (1 to k-1)
            if si_sub equals sj_sub:
                m[i,j] = 1
            else:
                m[i,j] = 0

return m
```

- Concatenate every possible combination of adjacent strings.
- Return a list of contiguous sequences *contigs*

for every string find the next closest strings and for each of these append it to the end of the previous

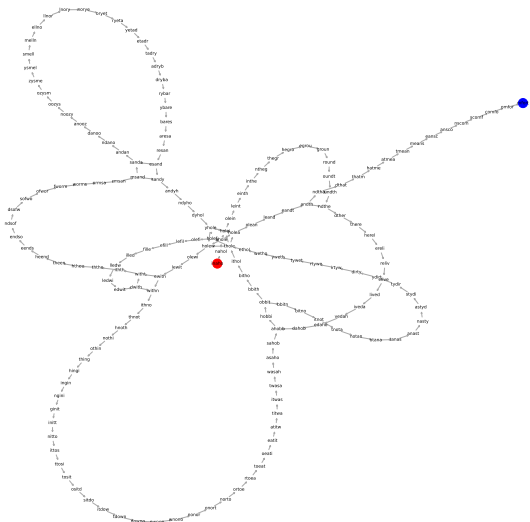


Figure 3: A graph of the 5-length kmers from the first sentence of The Hobbit.

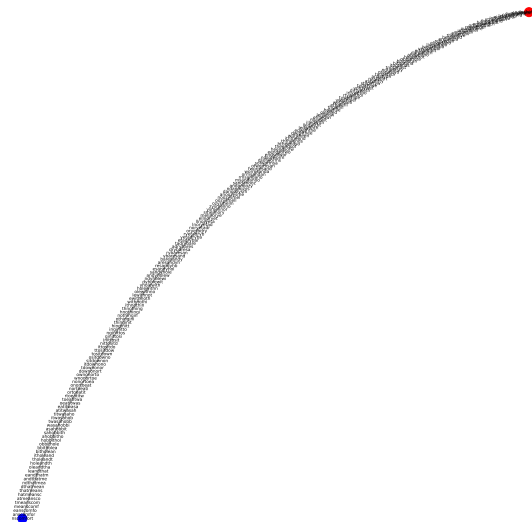


Figure 4: A graph of the 9-length kmers from the first sentence of The Hobbit.

5-letter versus 9-letter kmers



```
let S be as list of k-length strings
let A be an adjacency matrix between strings in S, with row and column names S, where rows encode outgoing and columns incoming edges

let colsums be the column sums of A
let starts be a list of strings of S where colsums equals the minimum value in colsums (S[colsums == min(colsums)])

contigs = []

define _recurse as a function of curr, contig, visited:
    row = A[ S==curr, : ] # ``
    nexts = S[row>0] # ``

    if len(nexts):
        contigs+=contig
        stop function execution

    for next in nexts:
        if next in visited:
            contigs+=contig
            continue to next loop
        new_visited = visited.copy() # ensure each branch only sees within branch visits
        new_visited[next] +=1

        _recurse(next, contig+=next[-1], new_visited) # recursively run function, each time appending contig with the last character of next

    for start in starts:
        let counter be a dictionary
        counter[start] = 1
        _recurse(start, start, counter)

return contigs
```

- Find the longest sequence contig that contains a query sequence.

```
let C be the list of contiguous strings  
let q be the query string
```

```
max_l = 0
```

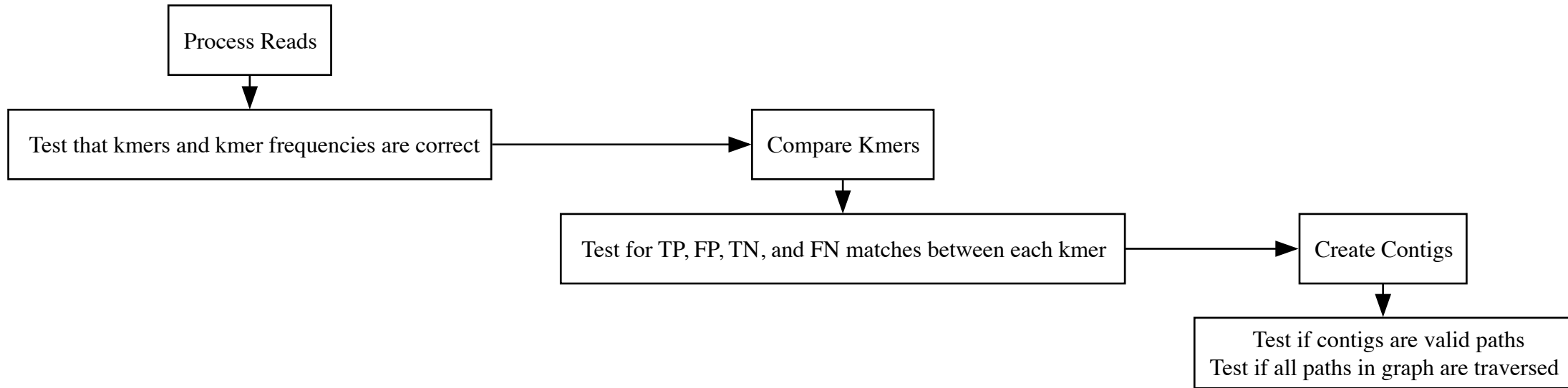
```
for c in C:  
    l = len(c)  
    if q is in c & l > max_l:  
        max_l = l  
        longest_with_query = c
```

```
return longest_with_query
```

Query: `dirtywetholefilledwiththeendsofwormsandanoozysmell`

Longest contigs with query sequence:

in a hole in the ground there lived a hobbit not a nasty `dirtywetholefilledwiththeendsofwormsandanoozysmell` nor yet a dry bares and
in a hole in the ground there lived a hobbit not a nasty `dirtywetholefilledwiththeendsofwormsandanoozysmell` nor yet a dry bares and y hole
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in a hole in the ground there lived a hobbit not a nasty `dirtywetholefilledwiththeendsofwormsandanoozysmell` nor yet a dry bares and y hole and that means comfort
in a hole with nothing in it to sit down on nor to eat it was a hobbit not a nasty `dirtywetholefilledwiththeendsofwormsandanoozysmell` nor yet a dry bares and
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REFERENCES

[1] J. R. R. Tolkien, *The Hobbit*. HarperCollins, 2012.

<https://github.com/canderson318/research-methodology-programming-assignment>